

Q6. Wild Thing

Command interfaces to modern operating systems often include a number of metacharacters which provide powerful features that allow a user to accomplish a lot of work without resorting to traditional programming. One of the most common metacharacters is a wildcard character.

Wildcard characters are used within filenames to permit the user to generically specify a group of files that a command should act upon. A wildcard character itself matches zero or more characters in the same position as the wildcard.

Suppose a user wanted to print all files ending in .c, delete all files beginning with old_ , and copy all files (no matter what they are named) to a floppy disk. With wildcard metacharacters, all of this can be done using only three commands instead of using a separate command for each file involved. The metacharacter sequences for these commands would be as follows:

*.c

old_*

*

Write a program that will accept an input pattern which may contain wildcard characters and decide whether or not each of the filenames that follow it matches the pattern

Input

The input file will contain a series of datasets. Each dataset will consist of a pattern line followed by a list of filenames to compare against the pattern. For the purposes of this program, a filename may consist of any group of one to twenty characters, not including asterisks. None of these characters should receive special treatment (periods, for example, should behave as any other characters matched).

The pattern line may include zero or more wildcard characters. A wildcard character will be represented by an asterisk (*).

The list of filenames will appear one per line following the pattern line. There will be an unknown number of filenames in the list.

All lines of input will be one to twenty characters in length, including the pattern line. The input file will contain an unknown number of datasets, each separated by a blank line.

Output

The output for this program will be a short report showing the pattern and the filenames from the subsequent list that match the pattern. Filenames that do not match the pattern should not be printed out. The results from each input dataset should be separated by one blank line.

See the example for the exact format.

Notice that 'MATCHES FOR PATTERN: HELLO' was not displayed, because there were no matches in its filename list.

Sample Input

C*AT

COMFILE.DAT

COST.DATA

CAT

%XCAT

COAT

CATCH

A

MOUNTAIN.TXT

ALFRED

PROG1A

SECOND.ED

PROG1A.PAS

HELLO

NOTHING

B**N*

NIBBLE.BIT

BANANA

BNXJ.25

BORN

ABNORMAL.LIS

BRANDISH.SRD

BITNET

Sample Output

MATCHES FOR THE PATTERN: C*AT

COMFILE.DAT

CAT

COAT

MATCHES FOR THE PATTERN: *A*

MOUNTAIN.TXT

ALFRED

PROG1A

PROG1A.PAS

MATCHES FOR THE PATTERN: B**N*

BANANA

BNXJ.25

BORN

BRANDISH.SRD

BITNET